

Data Preprocessing and Exploratory Data Analysis

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Outline

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- 2 Features and Feature Engineering Basics
- 3 Data Preprocessing and Cleaning
- 4 Exploratory Data Analysis (EDA) & Data Visualization

Data Collection and Representation

Data Collection and Representation

Data Sources & Collection

Data Sources

- Web APIs and web scraping
- Batch ingestion vs. real-time streaming pipelines
- Public ML repositories (Kaggle, UCI Machine Learning Repository)
- Administrative and survey microdata (e.g. DHS/MICS-style datasets), typically collected under a complex survey design with sampling weights

Data Types

- Tabular data
- Unstructured data: text, images, audio
- Time-series data
- Graph / network data
- Structured vs. semi-structured: relational tables vs. JSON-style API responses

Data Collection and Representation

Linear Algebra Foundations & Mathematical
Representation

Feature Vectors

- Each data point is represented as a d -dimensional feature vector.
- For the i -th instance in the dataset,

$$x^{(i)} = \begin{bmatrix} x_1^{(i)} & x_2^{(i)} & \dots & x_d^{(i)} \end{bmatrix}^T \in \mathbb{R}^d$$

- The superscript (i) indexes the instance; the subscript indexes the feature.

The Design Matrix

- Stacking n instances gives an $n \times d$ design matrix,

$$X = \begin{bmatrix} x^{(1)T} \\ x^{(2)T} \\ \vdots \\ x^{(n)T} \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1d} \\ x_{21} & x_{22} & \dots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nd} \end{bmatrix} \in \mathbb{R}^{n \times d}$$

- Row i of X is the transpose of instance i 's feature vector, $x^{(i)T}$; column j holds every instance's value for feature j .

Target Vector

- Target variables are collected into a vector $y \in \mathbb{R}^n$, one entry per instance.
- Continuous values for regression (e.g. price, temperature); discrete class labels for classification (e.g. spam / not spam).

Practical Scenario: Real Estate Price Prediction

- Each row of X is one house: $x^{(i)} = [x_{\text{sqft}}, x_{\text{beds}}, x_{\text{baths}}, x_{\text{zip}}]^T$.
- For three houses,

$$X = \begin{bmatrix} 1400 & 3 & 2 & 10001 \\ 2200 & 4 & 3 & 10002 \\ 950 & 2 & 1 & 10001 \end{bmatrix}, \quad y = \begin{bmatrix} 310 \\ 480 \\ 205 \end{bmatrix} \text{ (\$1000s)}$$

- The model learns a mapping from each row of X to the corresponding entry of y .

Features and Feature Engineering Basics

Features and Feature Engineering Basics

Feature Taxonomy

Feature Taxonomy

- **Categorical:** Nominal (unordered, e.g. color, region), Ordinal (ordered, e.g. education level)
- **Numerical:** Discrete (integer counts), Continuous (real numbers)
- **Binary:** Logical indicators, 0 or 1

Features and Feature Engineering Basics

Categorical Encoding Techniques

One-Hot Encoding (Dummy Variables)

- Expands a categorical column with k levels into k (or $k - 1$) binary indicator columns.
- Used for nominal data with distance- or gradient-based algorithms (Linear Regression, Neural Networks, SVMs), to avoid implying an ordinal relationship that doesn't exist.
- Example: `Color` $\in$ {Red, Blue, Green} becomes `Is_Red`, `Is_Blue`, `Is_Green`.

Ordinal / Label Encoding

- Maps ordered categories to an integer sequence, 0, 1, 2, ...
- Appropriate for genuinely ordinal features, or for tree-based algorithms (Random Forests, XGBoost) that split on integers efficiently without expanding dimensionality.
- Example: satisfaction responses [Poor, Satisfactory, Excellent] $\rightarrow$ [0, 1, 2].

Features and Feature Engineering Basics

Feature Scaling & Normalization

Standardization (Z - Score Scaling)

- Formula: $x' = \frac{x - \mu}{\sigma}$, giving $\mu = 0$, $\sigma = 1$.
- Needed for algorithms sensitive to scale and to gradient descent step size (KNN, Linear/Logistic Regression, SVMs, Neural Networks).
- Example: bringing Age (0–100 years) and Annual Income (\$10,000–\$1,000,000) onto a common scale.
- Fit μ and σ on the training set only, then apply to validation/test — the same leakage principle as target encoding.

Min - Max Scaling

- Formula: $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$, bounding data to $[0, 1]$.
- Used for non-Gaussian features needing a bounded range, or to preserve zero values in sparse matrices.
- Example: normalizing image pixel intensities (0–255) before neural network training.
- Terminology note: this and standardization are column-wise *scaling*. Row-wise *normalization* (e.g. rescaling each instance vector to unit L_2 norm) is a distinct operation — the two terms get conflated often.

Robust Scaling (Median & IQR)

- Formula: $x' = \frac{x - \text{Median}}{\text{IQR}}$.
- Used when scale-sensitive algorithms meet data with legitimate extreme values that would distort mean/variance-based scaling.
- Example: scaling executive compensation data with a heavy upper tail.

Data Preprocessing and Cleaning

Data Preprocessing and Cleaning

Handling Missing Values

Missingness Mechanisms

- **MCAR** (Missing Completely at Random): missingness is independent of both observed and unobserved data.
- **MAR** (Missing at Random): missingness depends on observed data, but not on the missing value itself.
- **MNAR** (Missing Not at Random): missingness depends on the unobserved value itself (e.g. high earners skipping the salary question).

Deletion Strategies

- **Listwise deletion:** drop entire rows with any missing entry.
Reasonable under MCAR with minimal missingness ($< 2-5\%$); risky under MAR/MNAR, where it introduces bias and shrinks the sample.
- **Pairwise deletion:** use all available data for each calculation (e.g. each pairwise correlation) rather than discarding a row entirely.
Avoids losing data but can leave summary statistics computed on inconsistent subsamples.

Simple Imputation

- **Mean / Median / Mode Imputation:** replace missing entries with a global summary statistic — median for numerical/skewed features, mode for categorical.
- Appropriate for baseline modeling with minimal MCAR missingness ($< 5\%$).

Model - Based Imputation

- **k -Nearest Neighbors (KNN) Imputation:** imputes using feature distance to neighboring rows. Suited to tabular data with moderate missingness (5–15%) where features are correlated.
- **MICE** (Multivariate Imputation by Chained Equations): models each variable with missing values as a function of the others, iteratively. Suited to high-stakes statistical/clinical modeling under MAR, with missingness spread across multiple features.
- Fit both on the training fold only, then apply to validation/test — the same leakage principle as target encoding.

Data Preprocessing and Cleaning

Handling Outliers

Outlier Detection: IQR and Z -Score

- **Tukey's IQR rule:** flag points outside $[Q1 - 1.5 \times \text{IQR}, Q3 + 1.5 \times \text{IQR}]$.
- **Z -score rule:** flag points where $|z| = \left| \frac{x - \mu}{\sigma} \right| > 3$ (or another chosen threshold).
- Z -score reuses the standardization formula from Section 2; it assumes roughly normal data and is sensitive to the same outliers it's trying to detect, since they inflate μ and σ — the IQR rule is more robust to that feedback loop.

Outlier Detection: Isolation Forest

- Model-based detection: measures how quickly an instance is isolated by random tree partitions — outliers isolate in fewer splits.
- Suited to high-dimensional tabular data, where single-feature rules (IQR, Z -score) miss multivariate anomalies.
- Example: credit card fraud detection, where a fraudulent transaction may look normal on any one feature but anomalous jointly.

Outlier Treatment

- **Trimming / culling:** remove verified, corrupt outlier rows.
- **Winsorization (capping):** replace extreme values with a boundary percentile (e.g. 1st/99th), preserving sample size while limiting the influence of tail values on parametric regression coefficients.

Data Preprocessing and Cleaning

Data Integrity & Hygiene

Data Integrity & Hygiene

- Deduplication of redundant rows
- Structural corrections: unit conversions, standardizing date-time strings, fixing casing errors

Exploratory Data Analysis (EDA) & Data Visualization

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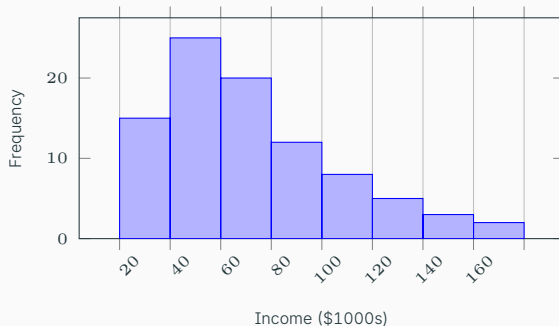
Summary Statistics & Central Tendency

Summary Statistics

- **Central tendency:** Mean, Median, Mode
- **Dispersion:** Variance, Standard Deviation, Interquartile Range (IQR)
- First step of EDA: establishes central values, spread, and baseline data health before any modeling

Distribution Shape

- Compare mean and median to detect skew: right-skewed income data shows Mean (\$85,000) well above Median (\$52,000).
- A long right tail (a few very high earners) pulls the mean up without moving the bulk of the distribution — visible directly in a histogram.

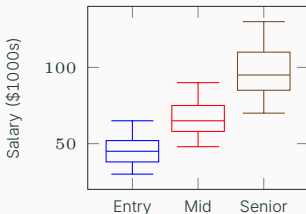


Exploratory Data Analysis (EDA) & Data Visualization

Groupwise Analysis & Segment Distributions

Groupwise Boxplots

- Compare a continuous feature's distribution across discrete groups: median, spread, and outliers per group, side by side.
- Example: salary distributions by job level, or recovery times by treatment arm (Placebo, Drug A, Drug B).
- Reveals sub-group variance and median shifts that a single overall summary statistic would hide.



Groupwise Aggregations & Contingency Tables

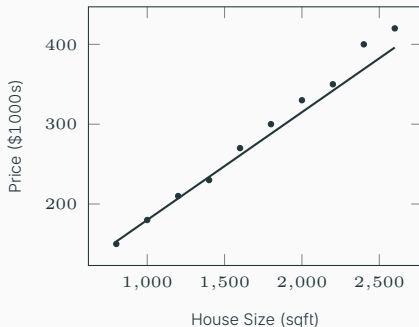
- Cross-tabulate a categorical outcome against a categorical predictor to segment behavior by group.
- Example: churn status against subscription tier (Basic, Pro, Enterprise), to find segment-specific churn rates.
- Watch for Simpson's Paradox: an association that holds within every subgroup can reverse when the subgroups are pooled.

Exploratory Data Analysis (EDA) & Data Visualization

Bivariate & Multivariate Structural Analysis

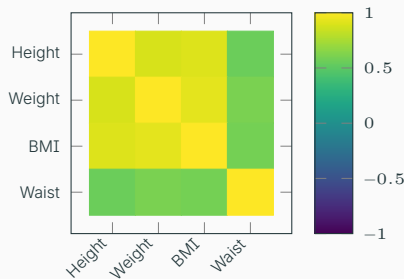
Scatter Plots & Trendlines

- Examine pairwise relationships between a continuous feature and the target.
- Example: House_Size vs. Sales_Price, with a fitted trendline to check linearity.



Correlation Matrices & Heatmaps

- Metrics: Pearson linear correlation (r), Spearman rank correlation.
- Screen features for multicollinearity before training a model.
- Example: pairwise correlations among Height, Weight, BMI, Waist_Circumference to flag collinear pairs ($r > 0.85$).



Questions?
